Gun-Woo Bae

Homepage   Tech Blog   Linkedin   Github

Education

Gwangju Institute of Science and Technology (GIST) **Feb. 2024 – Feb. 2026**
AIMED Lab of Prof. Mansu Kim *Gwangju, Korea*
Master of AI Convergence
• GPA: 3.91 / 4.5 (3.48/4.0)

Jeju National University **Feb. 2017 – Aug. 2023**
Bachelor of Computer Science and Statistics & History *Jeju, Korea*
• GPA: 4.22 / 4.5 (3.75/4.0)

Publications

† Equal contribution   ∗ Corresponding author

Gunwoo Bae†, Yeonwoo Kim†, Junha Bae, Mansu Kim* "ConnecToMind2: Inter-Subject fMRI Decoding via Whole-Brain Connectome-Guided Alignment", *MICCAI*, 2026. [Spotlight, Early Accept(9%)]

Youngjoo Lee†, **Gunwoo Bae**†, H. Suh, J. Cha, J. Jung, Park*, and Mansu Kim* "A real-world Fitbit-derived dataset of activity, sleep, and heart rate with matched clinical factors in on-treatment lung cancer patients", *Scientific Data*, 2026. [IF: 7.2, Top 11.8%] (Link) (Dataset)

Gunwoo Bae†, Yeonwoo Kim†, Mansu Kim* "ConnecToMind: Connectome-Aware fMRI Decoding for Visual Image Reconstruction", *MICCAI MLMI Workshop*, 2025. [Poster] (Link)

Gunwoo Bae, Mansu Kim*, "Image classification based on functional MRI signal", *Conference of the Korean Society of Medical and Biological Engineering*, 2024. [Oral, Bronze Award] (Link)

Research Experience

Research 1: Neural Decoding

- Developed a framework for reconstructing visual images from fMRI signals.
- Validated that the region-specific embedding preserved the functional roles of brain regions during visual reconstruction.
- Demonstrated that cortical regions outside the visual network also significantly influence visual stimuli reconstruction.

  Published as:

  – *"ConnecToMind2: Inter-Subject fMRI Decoding via Whole-Brain Connectome-Guided Alignment," MICCAI, 2026.*
  – *"ConnecToMind: Connectome-Aware fMRI Decoding for Visual Image Reconstruction," MICCAI MLMI Workshop, 2025.*

Research 2: Wearable Device Data Analysis with Samsung Medical Center

- Collected 3-month wearable data, including activity, sleep, and heart rate, from approximately 250 lung cancer patients.
- Validated the reliability of the dataset by comparing its trends with previous studies.
- Developed an automated data collection platform featuring interactive visualizations for clinicians and personalized reports for patients.

  Published as:

  – *"A real-world Fitbit-derived dataset of activity, sleep, and heart rate with matched clinical factors in on-treatment lung cancer patients," Scientific Data, 2026.*

Projects

AutoX: Cloud Platform for Resource Sharing **Jan. 2025 – Present**
In collaboration with GIST AI Convergence Department

- **Purpose:** Build a cloud service platform that enables sharing of idle GPU and CPU resources across research labs to support AI training tasks.
- **Role:** Connected Kubernetes with the back-end system, deployed the web service on Kubernetes, and implemented automated job scheduling and management using Airflow.
- **Tech Stack:** Kubernetes, Airflow, Django REST Framework, PostgreSQL, Vue.js, Nginx
- **Link:** www.autox.dreamai.kr

G-FITBIT: Platform to Monitor Patients Using Wearable Devices **Oct. 2024 – Feb. 2026**
In collaboration with Samsung Medical Center

- **Purpose:** Continuously monitor the health status of cancer patients using wearable devices and enable two-way feedback between patients and hospitals.
- **Role:** Built the back-end system, deployed the web platform, and automated data collection from wearable devices.
- **Tech Stack:** Django REST Framework, PostgreSQL, Airflow, Docker, Vue.js, Nginx
- **Link:** www.seouloncology.com

Technical Skills

Languages: Python, R, C

AI Tools: Pytorch, Numpy, Pandas

Server Tools: Kubernetes, Docker, Cuda, Airflow

Brain Analysis Tools: Freesurfer, ANTs, FSL, fMRIPrep

Web Tools: Django, Django REST Framework, PostgreSQL

Scholarships & Awards

Best Paper Award (Bronze) **Oct. 2024**
Conference of the Korean Society of Medical and Biological Engineering

GIST Full Scholarship (Full Funding) **Mar. 2024 – Feb. 2026**
National scholarship student of Korea

Academic Excellence Scholarship (Half Funding) **Aug. Feb. 2022, Aug. 2020**
Jeju National University

Extracurricular Activities

Google Machine Learning Bootcamp
Google Korea **Sep. 2022 – Nov. 2022**

Military Service
Republic of Korea Army **July 2018 – Feb. 2020**